

Frontend Code

Student Management System

HTML, CSS and JavaScript files placed inside the public folder.

public/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Management System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header><h1>Student Management System</h1></header>

  <main>
    <section class="card">
      <h2 id="formTitle">Add Student</h2>
      <form id="studentForm">
        <input name="name" placeholder="Full name" required>
        <input name="email" type="email" placeholder="Email" required>
        <input name="course" placeholder="Course (e.g. B.Tech CSE)" required>
        <input name="year" type="number" min="1" max="6" placeholder="Year" required>
        <input name="phone" placeholder="Phone">
        <div class="actions">
          <button type="submit">Save</button>
          <button type="button" id="cancelBtn" class="secondary" hidden>Cancel</button>
        </div>
      </form>
    </section>

    <section class="card">
      <input id="search" placeholder="Search by name or course...">
      <div class="table-wrap">
        <table>
          <thead>
            <tr>
              <th>ID</th><th>Name</th><th>Email</th><th>Course</th>
              <th>Year</th><th>Phone</th><th>Actions</th>
            </tr>
          </thead>
          <tbody id="tbody"></tbody>
        </table>
      </div>
    </section>
  </main>

  <script src="script.js"></script>
</body>
</html>
```

public/style.css

```
* { box-sizing: border-box; }

body {
  margin: 0;
  font-family: Arial, Helvetica, sans-serif;
  background: #f3f5f9;
  color: #222;
}

header {
  background: #2f4ea2;
```

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    color: #fff;
    padding: 16px 24px;
}
header h1 { margin: 0; font-size: 22px; }

main { max-width: 960px; margin: 24px auto; padding: 0 16px; }

.card {
  background: #fff;
  border-radius: 8px;
  padding: 18px;
  margin-bottom: 20px;
  box-shadow: 0 1px 4px rgba(0, 0, 0, 0.1);
}
.card h2 { margin-top: 0; font-size: 18px; }

form {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 10px;
}

input {
  padding: 9px 10px;
  border: 1px solid #c5cad6;
  border-radius: 6px;
  font-size: 14px;
}
#search { width: 100%; margin-bottom: 12px; }

button {
  padding: 9px 16px;
  border: none;
  border-radius: 6px;
  background: #2f4ea2;
  color: #fff;
  cursor: pointer;
  font-size: 14px;
}
button.secondary { background: #6b7280; }
button.danger { background: #d64545; }
button:hover { opacity: 0.9; }

.actions { display: flex; gap: 8px; }

.table-wrap { overflow-x: auto; }
table { width: 100%; border-collapse: collapse; }
th, td {
  padding: 10px;
  text-align: left;
  border-bottom: 1px solid #e5e7eb;
  font-size: 14px;
}
th { background: #eef1f8; }
td button { padding: 5px 10px; font-size: 12px; margin-right: 4px; }

```

public/script.js

```

const API = '/api/students';
const form = document.getElementById('studentForm');
const tbody = document.getElementById('tbody');
const searchBox = document.getElementById('search');
const formTitle = document.getElementById('formTitle');
const cancelBtn = document.getElementById('cancelBtn');

let students = [];
let editId = null;

// Escape text before putting it into HTML (prevents XSS)
const esc = (s) => String(s ?? '').replace(/[<>"]/g, (c) => (
  { '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;' }[c]
));

async function loadStudents() {
  const res = await fetch(API + '?search=' + encodeURIComponent(searchBox.value));

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students = await res.json();
tbody.innerHTML = students.map((s) => `
  <tr>
    <td>${s.id}</td><td>${esc(s.name)}</td><td>${esc(s.email)}</td>
    <td>${esc(s.course)}</td><td>${s.year}</td><td>${esc(s.phone)}</td>
    <td>
      <button onclick="editStudent(${s.id})">Edit</button>
      <button class="danger" onclick="deleteStudent(${s.id})">Delete</button>
    </td>
  </tr>`).join('');
}

form.addEventListener('submit', async (e) => {
  e.preventDefault();
  const body = Object.fromEntries(new FormData(form));
  const res = await fetch(editId ? `${API}/${editId}` : API, {
    method: editId ? 'PUT' : 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify(body)
  });
  if (!res.ok) {
    const data = await res.json();
    alert(data.error || 'Something went wrong');
    return;
  }
  resetForm();
  loadStudents();
});

function editStudent(id) {
  const s = students.find((x) => x.id === id);
  for (const key of ['name', 'email', 'course', 'year', 'phone']) {
    form.elements[key].value = s[key] ?? '';
  }
  editId = id;
  formTitle.textContent = 'Edit Student';
  cancelBtn.hidden = false;
  window.scrollTo({ top: 0, behavior: 'smooth' });
}

async function deleteStudent(id) {
  if (!confirm('Delete this student?')) return;
  await fetch(`${API}/${id}`, { method: 'DELETE' });
  loadStudents();
}

function resetForm() {
  form.reset();
  editId = null;
  formTitle.textContent = 'Add Student';
  cancelBtn.hidden = true;
}

cancelBtn.addEventListener('click', resetForm);
searchBox.addEventListener('input', loadStudents);

loadStudents();

```